

Pimpri Chinchwad Education Trust 's
PIMPRI CHINCHWAD COLLEGE OF ENGINEERING,NIGDI PUNE-44
 (An Autonomous Institute Affiliated to Savitribai Phule Pune University)

[BSH22BS07] MULTIVARIATE CALCULUS

Department: AS&H

Academic Year: 2025-26

Semester: II

Year	Course Name	Course Code	Credits	FA	SA	Total
F.Y.B.Tech	MULTIVARIATE CALCULUS	BSH22BS07	2	20 M	30 M	50 M

Number of Questions:

	Easy	Moderate	Difficult	Total
Unit-IV	25	25	25	75

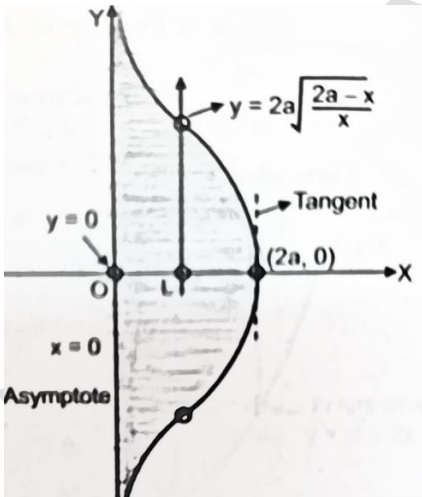
UNIT-IV: MUTIPLE INTEGRAL

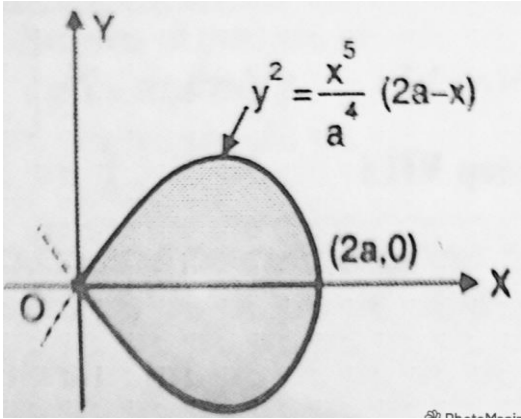
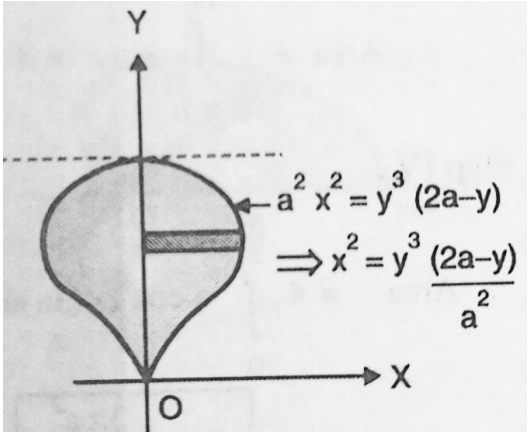
DOUBLE INTEGRAL

TYPE-I:DIRECT EVALUATION OF DOUBLE INTEGRAL		Ans	Marks	Level
1.	Evaluate $\int_0^1 \int_0^{\sqrt{1+x^2}} \frac{dx dy}{1+x^2+y^2}$	Ans: $\frac{\pi}{4} \log(1+\sqrt{2})$	4	Easy
2.	Evaluate $\int_0^a \int_0^{\sqrt{a^2-y^2}} \sqrt{a^2-x^2-y^2} dx dy$	Ans: $\frac{\pi a^3}{6}$	4	Medium
3.	Evaluate $\int_0^1 \int_0^1 \frac{dx dy}{\sqrt{(1-x^2)(1-y^2)}}$	Ans: $\frac{\pi^2}{4}$	5	Difficult
4.	Evaluate $\int_0^1 \int_0^x e^{y/x} dx dy$	Ans: $\frac{e-1}{2}$	4	Easy
5.	Evaluate $\int_0^1 \int_0^y xy e^{-x^2} dx dy$	Ans: $\frac{1}{4e}$	4	Easy
6.	Evaluate $\int_0^\infty \int_0^\infty x e^{-x^2(1+y^2)} dx dy$	Ans: $\frac{\pi}{4}$	4	Easy
TYPE-II: EVALUATION $\iint_R f(x,y) dx dy$ IF THE LIMITS ARE		Ans	Marks	Level

NOT GIVEN :				
1.	Evaluate $\iint_R xy(x+y)dxdy$ over the area bounded by $y = x^2$ and $y = x$	Ans: $\frac{3}{56}$	4	Medium
2.	Evaluate $\iint_R y dxdy$ over the area bounded by $y = x^2$ and $x + y = 2$	Ans: $\frac{36}{5}$	4	Easy
3.	Evaluate $\iint_R (x^2 + y^2)dxdy$ over the area of triangle whose vertices are (0,1) (1,1) and (1,2)	Ans: $\frac{7}{6}$	4	Medium
4.	Evaluate $\iint_R e^{y^2} dxdy$ over the area of triangle whose vertices are (0,0) (2,1) and (0,1)	Ans: $e-1$	4	Easy
5.	Evaluate $\iint_R \sqrt{4x^2 - y^2} dxdy$ over the area of the region bounded by $y = 0$, $y = x$ and $x = 1$	Ans: $\frac{1}{3} \left(\frac{\pi}{3} + \frac{\sqrt{3}}{2} \right)$	4	Medium
6.	Evaluate $\iint_R xy dxdy$ over the region bounded by X-axis, ordinate at $x = 2a$ and the parabola $x^2 = 4ay$	Ans: $\frac{a^4}{3}$	4	Easy
7.	Solve $\iint_R xy dxdy$ over the region bounded by the $x^2 = y$ and $y^2 = -x$	Ans: $-\frac{1}{12}$	4	Easy
8.	Evaluate $\iint_R \sqrt{xy(1-x-y)} dxdy$ over the region bounded by $x = 0$, $y = 0$ and $x + y = 1$.	Ans: $\frac{2\pi}{105}$	5	Difficult
9.	Evaluate $\iint_R \frac{dxdy}{x^4 + y^2}$ over the region by $y \geq x^2$, $x \geq 1$	Ans: $\frac{\pi}{4}$	4	Medium
10.	Evaluate $\iint_R \frac{xy dxdy}{\sqrt{1-y^2}}$ over the positive quadrant of the circle $x^2 + y^2 = 1$.	Ans: $\frac{1}{6}$	5	Difficult
TYPE-III: TRANSFORM OF DOUBLE INTEGRAL INTO POLAR: (LIMIT NOT GIVEN)		Ans	Marks	Level
1.	Evaluate $\iint_R \sin(x^2 + y^2) dxdy$ over circle $x^2 + y^2 = a^2$	Ans: $\pi(1 - \cos a^2)$	4	Easy
2.	Evaluate $\iint_R \frac{x^2 y^2}{x^2 + y^2} dxdy$ where R is annulus region between the circle $x^2 + y^2 = 4$ and $x^2 + y^2 = 9$	Ans: $\frac{65\pi}{16}$	4	Medium

3.	Evaluate $\iint_R y^2 dx dy$ over the area which lies outside the circle $x^2 + y^2 = ax$ and inside the circle $x^2 + y^2 = 2ax$	Ans: $\frac{15a^4\pi}{64}$	4	Medium
4.	Evaluate $\iint_R xy \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} \right)^{\frac{1}{2}} dx dy$ where R is the area of ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ in positive quadrant	Ans: $\frac{a^2b^2}{5}$	5	Difficult
TYPE-IV: TRANSFORM OF DOUBLE INTEGRAL INTO POLAR: (LIMIT ARE GIVEN)		Ans	Marks	Level
1.	Evaluate $\int_0^a \int_0^{\sqrt{a^2-x^2}} e^{-(x^2+y^2)} dx dy$	Ans: $\frac{\pi}{4}(1-e^{-a^2})$	4	Easy
2.	Evaluate $\int_0^a \int_0^{\sqrt{a^2-x^2}} \sin\left(\frac{\pi}{a^2}(a^2-x^2-y^2)\right) dx dy$	Ans: $\frac{a^2}{2}$	4	Medium
3.	Evaluate $\int_0^1 \int_{\sqrt{x-x^2}}^{\sqrt{1-x^2}} \frac{xye^{-(x^2+y^2)}}{(x^2+y^2)} dx dy$	Ans: $\frac{1}{4}\left(1-\frac{2}{e}\right)$	5	Difficult
4.	Evaluate $\int_0^a \int_{\sqrt{ax-x^2}}^{\sqrt{a^2-x^2}} x\sqrt{x^2+y^2} dx dy$	Ans: $\frac{7a^4}{60}$	4	Medium
5.	Evaluate $\int_0^{\frac{a}{\sqrt{2}}} \int_x^{\sqrt{a^2-x^2}} \frac{x}{x^2+y^2} dx dy$	Ans: $a\left(1-\frac{1}{\sqrt{2}}\right)$	4	Easy
6.	Evaluate $\int_0^{\frac{a}{\sqrt{2}}} \int_y^{\sqrt{a^2-x^2}} \log(x^2+y^2) dx dy$	Ans: $\frac{\pi a^2}{8}(2\log a - 1)$	4	Medium
7.	Evaluate $\int_0^2 \int_0^{\sqrt{4-y^2}} \frac{y}{\sqrt{4-x^2}\sqrt{x^2+y^2}} dx dy$	Ans: π	5	Difficult
8.	Evaluate $\int_0^{2a} \int_0^{\sqrt{2ax-x^2}} (x^2+y^2) dx dy$	Ans: $\frac{3\pi a^4}{4}$	4	Easy
TYPE-V: AREA		Ans	Marks	Level

1.	Find the area cardioids $r = a(1 + \cos \theta)$	Ans: $\frac{3\pi a^2}{2}$	4	Easy
2.	Find the area inside the circle $r = a \sin \theta$ and outside the cardioids $r = a(1 - \cos \theta)$	Ans: $\frac{a^2}{4}(4 - \pi)$	5	Difficult
3.	Find the total area included between the two cardioids $r = a(1 + \cos \theta)$ and $r = a(1 - \cos \theta)$	Ans: $2a^2\left(\frac{3\pi}{4} - 2\right)$	4	Easy
4.	Find the total area outside the circle $x^2 + y^2 = a^2$ and inside the cardioids $r = a(1 + \cos \theta)$	Ans: $a^2\left(2 + \frac{\pi}{4}\right)$	4	Medium
5.	Find the area common to the circle $x^2 + y^2 = 9$ and $x^2 + y^2 = 6x$	Ans: $6\pi - \frac{9\sqrt{3}}{2}$	5	Difficult
6.	Find the area bounded by the parabola $y = x^2$ and the line $y = 2x + 3$	Ans: $\frac{32}{3}$	4	Easy
7.	Find the area bounded by the parabola $y^2 = 4x$ and the line $2x - 3y + 4 = 0$	Ans: $\frac{1}{3}$	4	Easy
8.	Find the area enclosed between the parabola $y^2 = 4ax$ and $x^2 = 4ay$	Ans: $\frac{16a^2}{3}$	4	Medium
9.	Find the area between the curve $xy^2 = 4a^2(2a - x)$ and Y -axis . Refer following Fig.	Ans: $4\pi a^2$	5	Difficult
				
10.	Find the area bounded by the curve $a^4 y^2 = x^5(2a - x)$. Refer	Ans: $\frac{5}{4}\pi a^4$	5	Difficult

	following Fig 			
11.	Find the area bounded by the curve $a^2 x^2 = y^3 (2a - y)$. Refer following Fig 	Ans: π^2	5	Difficult
<u>TRIPLE INTEGRATION</u>				
TYPE-VI: TRIPLE INTEGRATION WITH LIMIT GIVEN:		Ans	Marks	Level
1.	Evaluate $\int_{-1}^1 \int_0^z \int_{x-z}^{x+z} (x + y + z) dx dy dz$	Ans: 0	4	Easy
2.	Evaluate $\int_0^1 \int_0^1 \int_{x-z}^1 e^{x+y+z} dx dy dz$	Ans: $(e-1)^3$	4	Medium
3.	Evaluate $\int_0^1 \int_{y^2}^1 \int_0^{1-x} x dx dy dz$	Ans: $\frac{4}{35}$	4	Easy

4.	Evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} \frac{dx dy dz}{\sqrt{1-x^2-y^2-z^2}}$	Ans: $\frac{\pi^2}{8}$	4	Medium
5.	Evaluate $\int_0^1 \int_0^{1-x} \int_0^{x+y} e^z dx dy dz$	Ans: $\frac{1}{2}$	4	Easy
TYPE-VII: TRIPLE INTEGRATION LIMIT NOT GIVEN:		Ans	Marks	Level
1.	Evaluate $\iiint_V x^2 y z dx dy dz$ throughout the volume bounded by the planes $x=0, y=0, z=0, \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.	Ans: $\frac{a^3 b^2 c^2}{2520}$	4	Easy
2.	Evaluate $\iiint_V (x+y+z) dx dy dz$ over the volume the tetrahedron bounded by the planes $x=0, y=0, z=0$, and $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$	Ans: $\frac{1}{8}$	4	Easy
3.	Evaluate $\iiint_V (x^2 y^2 + y^2 z^2 + x^2 z^2) dx dy dz$ throughout the volume of the sphere $x^2 + y^2 + z^2 = a^2$	Ans: $\frac{4\pi a^7}{35}$	4	Medium
4.	Evaluate $\iiint_V \frac{dx dy dz}{(1+x+y+z)^3}$ throughout the volume of tetrahedron bounded by the planes $x=0, y=0, z=0, x+y+z=1$	Ans: $\frac{1}{2} \left(\log 2 - \frac{5}{8} \right)$	5	Difficult
5.	Evaluate $\iiint_V \sqrt{x^2 + y^2} dx dy dz$, where V is volume bounded by surface $x^2 + y^2 = z^2, z > 0$ and $z=1$.	Ans: $\frac{\pi}{6}$	5	Difficult
6.	Evaluate $\iiint_V z^2 dx dy dz$ over the volume bounded by the cylinder $x^2 + y^2 = a^2$ and the elliptical paraboloid $x^2 + y^2 = z$ and the plane $z=0$	Ans: $\frac{\pi a^8}{12}$	5	Difficult
7.	Evaluate $\iiint_V z^2 dx dy dz$ over the volume common to sphere $x^2 + y^2 + z^2 = a^2$ and cylinder $x^2 + y^2 = ax$	Ans: $\frac{2\pi a^5}{15}$	4	Medium
8.	Evaluate $\iiint_V (x^2 + y^2) dx dy dz$ over the region bounded by the paraboloid $x^2 + y^2 = 3z$, and the plane $z=3$	Ans: $\frac{81\pi}{2}$	4	Easy

TYPE-VIII: TRIPLE INTEGRATION TRANSFORM TO SPHERICAL POLAR FORM :		Ans	Marks	Level
1.	Evaluate $\iiint_V \frac{dxdydz}{(1+x^2+y^2+z^2)^2}$ over the entire positive octant of the space.	Ans: $\frac{\pi^2}{8}$	4	Medium
2.	Evaluate $\iiint_V \frac{z^2}{x^2+y^2+z^2} dxdydz$ where V is the volume bounded by the sphere $x^2+y^2+z^2=z$	Ans: $\frac{\pi}{9}$	5	Difficult
3.	Evaluate $\iiint_V xyz dxdydz$ over the volume bounded by the sphere $x^2+y^2+z^2=4$ in the positive octant.	Ans: $\frac{4}{3}$	4	Easy
4.	Evaluate $\iiint_V \sqrt{1-\frac{x^2}{a^2}-\frac{y^2}{b^2}-\frac{z^2}{c^2}} dxdydz$ throughout the volume of ellipsoid $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$	Ans: $\frac{abc \pi^2}{4}$	4	Medium
5.	Evaluate $\iiint_V \frac{dxdydz}{\sqrt{1-x^2-y^2-z^2}}$ over the volume bounded by the sphere $x^2+y^2+z^2=1$ in the positive octant.	Ans: $\frac{\pi^2}{8}$	4	Medium
6.	Evaluate $\iiint_V \frac{dxdydz}{(x^2+y^2+z^2)^{3/2}}$ where V is the volume bounded by the sphere $x^2+y^2+z^2=a^2$ and $x^2+y^2+z^2=b^2$, ($b > a$)	Ans: $4\pi \log\left(\frac{b}{a}\right)$	5	Difficult
TYPE-IX: VOLUME		Ans	Marks	Level
1.	Find the volume of paraboloid of revolution $x^2+y^2=4z$ cut off by the plane $z=4$	Ans: 32π	4	Easy
2.	Find the volume of the region bounded by $z=x^2+y^2$ and $z=2x$.	Ans: $\frac{\pi}{2}$	4	Easy
3.	Find the volume of the region enclosed by the cone $z=\sqrt{x^2+y^2}$ and paraboloid $z=x^2+y^2$	Ans: $\frac{\pi}{6}$	5	Difficult
4.	Find the volume enclosed between the cylinders $x^2+y^2=2ax$ and $z^2=2ax$.	Ans: $\frac{128a^3}{15}$	5	Difficult
5.	Find the volume common to the cylinders $x^2+y^2=a^2$, $x^2+z^2=a^2$	Ans: $\frac{16a^3}{3}$	5	Difficult

6.	Find the volume of the cylinder $x^2 + y^2 = 2ax$, intercepted between the paraboloid $x^2 + y^2 = 2az$ and the XY -plane.	Ans: $\frac{3\pi a^3}{4}$	4	Medium
7.	Find the volume bounded by the paraboloid $x^2 + y^2 = az$ the cylinder $x^2 + y^2 = 2ay$ and the plane $z = 0$.	Ans: $\frac{3\pi a^3}{2}$	4	Medium
8.	Find the volume of the solid cut off from the sphere $x^2 + y^2 + z^2 = a^2$ by the cylinder $x^2 + y^2 = ax$.	Ans: $\frac{2\pi a^3}{3}$	5	Difficult
9.	Find the volume of the solid cut off from the sphere $x^2 + y^2 + z^2 = a^2$ by the cone $x^2 + y^2 = z^2$.	Ans: $\frac{2\pi a^3}{3} \left(1 - \frac{1}{\sqrt{2}}\right)$	4	Medium
10.	Find the volume bounded by the sphere $x^2 + y^2 + z^2 = 4$ and the paraboloid $x^2 + y^2 = 3z$.	Ans: $\frac{9\pi}{6}$	5	Difficult
11.	Find the volume bounded by the cylinder $x^2 + y^2 = 4$ and the planes $y + z = 4$ and $z = 0$	Ans: 16π	4	Medium
12.	Find the volume bounded by the cylinder $y^2 = x$, $x^2 = y$ and the planes $z = 0$, $x + y + z = 2$	Ans: $\frac{11}{30}$	5	Difficult
13.	Find the volume cut off from the paraboloid $x^2 + \frac{y^2}{4} + z = 1$ by the plane $z = 0$.	Ans: π	5	Difficult
15.	Find the volume of the tetrahedron bounded by the co-ordinate planes and the plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.	Ans: $\frac{abc}{6}$	4	Medium
16.	Find the volume of region bounded by paraboloid $x^2 + y^2 = 2z$ and cylinder $x^2 + y^2 = 4$.	Ans: 4π	5	Difficult
17.	Find the volume of region bounded by $z = 4 - x^2 - y^2$ the XY -plane.	Ans: 8π	5	Difficult
